🔥 UNIT I – Business Intelligence (10–12 MUST-DO QUESTIONS)

These **appear directly or indirectly every year**.

1. Explain **Business Intelligence** and its importance in decision making.
2. Explain **effective and timely decisions** with reference to BI.
3. Differentiate between **Data, Information and Knowledge** with examples.
4. Explain the **role of mathematical models in Business Intelligence**.
5. Draw and explain **Business Intelligence Architecture**. *(VERY frequent)*
6. Explain **Decision Support System (DSS)** and its main components.
7. Explain the **decision-making process** in organizations.
8. Explain the **Business Intelligence cycle / BI analysis cycle**.
9. Describe the **development phases of a Business Intelligence system**.
10. Explain **Ethics and ethical issues in Business Intelligence**.
11. Explain **data sources, data warehouse and data marts** in BI.
12. Short note: **BI methodologies** (OLAP, data mining, optimization, time series).

📌 **Exam tip:**
Architecture + DSS + Data→Information→Knowledge = **guaranteed marks**

🔥 UNIT II – Mathematical Models & Data Mining (10–12 MUST-DO QUESTIONS)

This unit is **very predictable**.

1. What is a **mathematical model**? Explain its structure.
2. Explain the **steps in developing a mathematical model**. *(Asked directly)*
3. Explain the **classes/types of mathematical models**.
4. Define **Data Mining** and explain its purpose.
5. Explain the **data mining process**.
6. What is **data preparation**? Explain its importance.
7. Explain **data validation** in data preparation.
8. Explain **data transformation** and why it is required.
9. What is **data reduction**? Explain with examples.

10. How does **data preparation improve accuracy of analysis**? *(Repeated)*
11. Explain **common methods used for data analysis in data mining**.
12. Short note: **Input data representation in data mining**.

📌 **Exam tip:**
Steps + definitions + "why important" questions are examiner favourites.

🔥 **UNIT III – Classification & Clustering (10–12 MUST-DO QUESTIONS)**

Almost **every question is repeated conceptually**.

Classification

1. What is **classification**? Explain classification problems.
2. Explain **Bayesian classification methods** and their purpose.
3. Explain **Logistic Regression** in classification.
4. Explain **Neural Networks** for classification.
5. Explain **Support Vector Machines (SVM)** and their role in classification.
6. Explain **evaluation of classification models**.

Clustering

7. What is **clustering**? How is it different from classification? *(Very common)*
8. Explain **clustering methods**.
9. Explain **partition-based clustering methods**.
10. Explain **hierarchical clustering methods**.
11. Why is **evaluation of clustering models** important?
12. Explain **one real-life application of clustering**.

📌 **Exam tip:**
Difference questions + "role of" + "why evaluation important" = safe bets.